

NEELIMA KODEDHALA

✉ neelcodedhala2002@gmail.com | ☎ +1 940-629-6602 | 🔗 LinkedIn | 📍 United States of America

Summary

Experienced Data Scientist and Software Engineer with over 2 years of hands-on experience in machine learning, data engineering, and software development. Skilled in developing and deploying predictive models using Scikit-learn, XGBoost, and Hugging Face's BERT, and proficient in building real-time data pipelines with Apache Kafka, PySpark, and AWS. Adept at working with large-scale data, optimizing ETL workflows, and creating interactive dashboards using Tableau and Power BI to support business decision-making. Experienced in applying Explainable AI techniques, anomaly detection, and time series forecasting to drive actionable insights. Strong foundation in cloud technologies (AWS, Azure), data visualization, and AI-driven decision-making, with a collaborative mindset and a commitment to delivering impactful solutions that drive business growth and efficiency.

Experience

Machine Learning Intern – Vosyn

July 2025 – Present

- Built an end-to-end MLOps pipeline for a voice synthesis model using Airflow, MLflow, and Docker, reducing training and deployment time by 40%.
- Helped with data preprocessing and audio cleaning using Librosa, improving model accuracy and audio quality by 25%.
- Set up CI/CD pipelines with GitHub Actions and AWS SageMaker for easy model updates and monitoring.
- Used Prometheus and Grafana to track model performance and resource usage, cutting issue resolution time by 30%.
- Fine-tuned Tacotron2 and WaveGlow models to make the generated voice sound more natural, improving quality scores from 3.6 to 4.4.
- Documented all steps and best practices to help future interns understand and maintain the system easily.

Graduate Research Assistant – University of North Texas

Jan 2024 – May 2025

- Developed machine learning and deep learning models (Logistic Regression, Random Forest, CNN, Transformer-based models) to predict healthcare outcomes, achieving 87% accuracy and 0.91 ROC-AUC on MIMIC-III and NIH Chest X-ray datasets.
- Applied Explainable AI (XAI) techniques (SHAP, LIME, Grad-CAM) to improve model interpretability, increasing clinician interpretability scores by 40% in faculty-led evaluations.
- Conducted comparative analysis of model performance, reducing false positives by 22% while maintaining high recall for critical patient cases.
- Developed an interactive Streamlit web app that reduced prediction-to-interpretation latency from minutes to seconds, enabling near real-time decision support.
- Delivered technical documentation and reproducible pipelines, supporting faculty and graduate peers in extending the system for future healthcare AI research.

Software Engineer – TCS

Apr 2022 – May 2023

- Curated and processed large-scale user-generated text and images by automating data cleaning, tokenization, and augmentation pipelines, reducing data preparation time by 30%.
- Designed and trained deep learning models for sentiment analysis and image classification, optimizing development cycles and reducing model training time by 28%.
- Integrated automated content generation workflows, streamlining manual curation efforts by 40% and enhancing content personalization aligned with sentiment outputs.
- Deployed scalable, real-time inference services with low-latency pipelines, improving system throughput by 35% and enabling high-volume content processing.
- Established monitoring systems and feedback loops that reduced misclassification incidents by 20% through continuous learning and adaptive model retraining.
- Developed interactive dashboards to visualize sentiment trends, content performance metrics, and user engagement KPIs, accelerating business decision-making.
- Automated model lifecycle workflows, reducing release cycles by 60% and improving infrastructure efficiency.

Technical Skills

Programming & Scripting: Python, SQL, R, JavaScript
Database & Storage: MySQL, PostgreSQL, MongoDB (NoSQL)
Data Visualization & BI: Power BI, Tableau, Looker Studio, MS Excel, Matplotlib, Seaborn
Machine Learning & AI: TensorFlow, PyTorch, Scikit-learn, XGBoost, Reinforcement Learning, Generative AI, GANs, Neural Networks (CNNs, RNNs)
Data Engineering & ETL: HDFS, Apache Spark, Hadoop, AWS Kinesis, Kafka, Data Warehousing, ETL Pipelines, Delta Lake, SQL-based ETL Processing
Advanced Analytics: Time Series Forecasting, Anomaly Detection, Risk Modeling, Customer Segmentation, Predictive Analytics, AI-Driven Decision Making
NLP & Text Analytics: Sentiment Analysis, Transformer Models, LangChain, Retrieval-Augmented Generation (RAG) Models
Cloud & Deployment: AWS SageMaker, AWS Lambda, AWS EC2, Docker, Kubernetes, Flask, Fast API, REST APIs, CI/CD, Azure ML, GCP Vertex AI
Soft skills: Problem Solving, Team Collaboration, Communication, Adaptability, Continuous Learning, Innovation

Projects

Analyzing Complex Sentences to Generate Images (2023):
Developed a web-based platform that generates high-quality, photo-realistic images from complex sentences using the Stable Diffusion model. The model applies a diffusion process to enhance image diversity and accuracy. Achieved 82% accuracy in capturing key elements and 85% reliability in maintaining visual appeal. *Published and presented this work at IOSR.*

Identifying Important Lines in a Research Paper (Research Project) (2022):
Developed a research-driven project using advanced NLP and machine learning algorithms to efficiently identify key lines in research papers. The intelligent model automates information extraction, reducing manual review time by 40%, improving content summarization by 30%, and significantly enhancing research productivity, accuracy, and overall efficiency by 25%.

Uninhibited (NGO)-Code for Good JP Morgan India Hackthon (2021):
Developed an interactive website for Uninhibited NGO in a hackathon to help expand its reach. Optimized the front-end, reducing load time by 25%, and strengthened security protocols, enhancing data protection by 30% and improved overall platform effectiveness.

Education

University of North Texas	Aug 2023 – May 2025
Master of Science in Computer Science	

Certifications

- AWS Certified Developer – Associate
- CS50 - Introduction to Programming in Python (Harvard)
- Generative AI Fundamentals (Google Cloud)